

100 Days of Azure DevOps

Day 23 Handout — trigger, pr, stages, jobs, steps — Lego instructions that care about whitespace

3 - Continuous Integration · Personal lab only · Educational content · Not a sales pitch

Architecture A — YAML schema map (outermost key to leaf)

Key	Meaning	Lab value
trigger	CI on push to listed branches	branches.include: [main]
pr	Validation on PRs targeting listed branches	branches.include: [main]
stages	Ordered groups; default one implicit stage	Build then Test
jobs	Agent allocation unit; can dependOn	BuildJob, TestJob
pool	vmImage or name	ubuntu-latest on each job
steps	script, bash, pwsh, task, checkout, template	echo is enough today
dependsOn	Stage/job graph	Test dependsOn Build
variables	compile-time vs runtime (Day 29)	Skip secrets today

Architecture B — decide stage vs job vs step

Need	Use	Do not
Different agent SKU / approval later	stage + job pool	One 40-step job named Deploy
Share a workspace / checkout	steps in the SAME job	dependsOn across jobs and expect files to be there
Fail-fast compile then tests	stage Test dependsOn Build	continueOnError: true on Build 'so Test can run'
PR only vs main only	pr: and trigger: separately	Assume trigger covers PRs — it does not
Reuse	template: later	Copy-paste 8 YAML files by hand
Logs as the product	displayName on every step	Untitled script steps you cannot find

Architecture C — pitfalls that skip PR validation or lose the workspace

Pitfall	What you see	Fix
trigger: none accidentally	Manual run only	Restore trigger.branches.include
No pr: key	PR has no Checks; merge is opinion	Add pr: branches include main; then branch policy
Job2 needs Job1 binaries	file not found on the second agent	PublishPipelineArtifact@1 / download, or same job
Tab vs space	Mapping values are not allowed here	Spaces only; VS Code YAML + schema
pool only at pipeline root, job overrides missing	One job silently on Default	Set pool on each job while learning
Skipped logs	You merge because the badge is green	Open the Test stage log end-to-end once

One-liner to remember

YAML pipelines are Lego instructions written by someone who enjoys whitespace arguments.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal azure-100-labs. Expand Day 21 hello into stages. Add a PR trigger. Read the logs. Hosted ubuntu-latest only.

- 1 Edit azure-pipelines.yml (or pipelines/ci.yml and retarget the pipeline) on feature/day23-yaml.
- 2 Add trigger.branches.include: [main] AND pr.branches.include: [main].
- 3 Add stages: Build (echo Building) then Test dependsOn: Build (echo Testing). Each job pool: vmImage: ubuntu-latest.
- 4 Push and open a PR to main. Confirm the pipeline appears on the PR Checks tab.
- 5 Pipelines → latest run → expand both stages. Read every step log. Note stage names in the graph.
- 6 If policies require a work item, link one. Squash merge when green.
- 7 Do not add a self-hosted pool name. Do not add secrets.

Two-stage YAML with CI + PR triggers

```
trigger:
  branches:
    include:
      - main
pr:
  branches:
    include:
      - main

stages:
- stage: Build
  jobs:
  - job: BuildJob
    pool:
      vmImage: ubuntu-latest
    steps:
    - script: echo Building...
      displayName: Compile stub

- stage: Test
  dependsOn: Build
  jobs:
  - job: TestJob
    pool:
      vmImage: ubuntu-latest
    steps:
    - script: echo Testing...
      displayName: Test stub
```

Done checklist

- Can name trigger, pr, stage, job, step, dependsOn
- PR to main queued a validation run
- Test stage ran after Build in the logs
- Both jobs used ubuntu-latest

Tomorrow — Day 24

CI pipeline for a .NET app